



A robot hand-eye calibration method of line laser sensor based on 3D reconstruction

Mingyang Li, Zhijiang Du, Xiaoxing Ma, Wei Dong, Yongzhuo Gao *

State Key Laboratory of Robotics and System, Harbin Institute of Technology (HIT), Harbin, China

ARTICLE INFO

Keywords:

Hand-eye calibration
Robot measurement
3D reconstruction
Point cloud processing
Optimization method

ABSTRACT

In the robotic eye-in-hand measurement system, a hand-eye calibration method is essential. From the perspective of 3D reconstruction, this paper first analyzes the influence of the line laser sensor hand-eye calibration error on the 3D reconstructed point clouds error. Based on this, considering the influence of line laser sensor measurement errors and the need for high efficiency and convenience in robotic manufacturing systems, this paper proposes a 3D reconstruction-based robot line laser hand-eye calibration method. In this method, combined with the point cloud registration technique, the newly defined error-index more intuitively reflects the calibration result than traditional methods. To raise the performance of the calibration algorithm, a Particle Swarm Optimization - Gaussian Process (PSO-GP) method is adopted to improve the efficiency of the calibration. The experiments show that the Root Mean Square Error (RMSE) of the reconstructed point cloud can reach 0.1256 mm when using the proposed method, and the reprojection error is superior to those using traditional hand-eye calibration methods.

1. Introduction

In the robotic manufacturing system, the in-situ measurement method of the robot is widely used due to the high efficiency and no need to change the fixture. The method of fixing the sensor to the end of the robot is called “eye-in-hand” style. This method benefits from the flexibility of the robot. Compared with a sensor fixed in one place, it is able to complete a broader range of measurements. At present, this measurement method is widely used in various types of robotic manufacturing systems, typically used in robot welding, grinding, assembly, and so on. The processed workpieces also cover a wide range of fields, including aerospace, automotive, energy, and so on [1,2].

As a relatively mature solution, robot combined with monocular cameras is currently widely used in the robotic measurement systems. The measurement result of this method is a two-dimensional image. With the help of the image processing algorithm and the internal parameters of the camera, it is usually possible to directly obtain the position of the measured object. However, the disadvantage of this method is that the accuracy is relatively low, and the information contained in two-dimensional images is insufficient, so it is difficult to accurately obtain the features in relatively complex tasks such as detecting freeform surfaces. In contrast, 3-Dimensional (3D) reconstruction of processed objects has higher accuracy and more intuitive data, so it is currently widely studied in different applications [3,4]. 3D

reconstruction is the establishment of mathematical models which is suitable for computer representation and processing. It is the basis for processing, manipulating, and analyzing the properties in a computer environment. The results of 3D reconstruction are usually in the form of point clouds, depth maps, or triangular meshes, which contain more accurate and intuitive information. For robotic manufacturing systems, such measurement results can be used for complex tasks. The methods of 3D measurement and reconstruction are generally divided into active methods and passive methods. Active methods include structured light, laser time-of-flight (TOF), and triangulation. Passive methods mainly include monocular vision and binocular/multiple vision methods. Considering measurement accuracy, working efficiency, and cost, some inspection products based on the triangular ranging method have certain advantages in robotic manufacturing systems. Among them, the line laser sensor has the advantages of high detection accuracy, small environmental impact, and relatively low cost, so it is widely used, and there already exist some mature commercial products [5].

The single measurement result of the line laser sensor is the X/Z coordinate of the laser beam irradiation position. When the line laser sensor is installed at the end of the robot, we can reconstruct objects in the form of point clouds with multiple scanning results by combining the corresponding robot poses [6]. However, in this “eye-in-hand” system, achieving accurate reconstructions requires precise calibration,

* Corresponding author.

E-mail address: gaoyongzhuo@hit.edu.cn (Y. Gao).

and among different types of calibrations, the hand-eye calibration is essential. Hand-eye calibration aims to obtain the relative transformation relationship between the end coordinate system of the robot (usually the center of the end flange) and the sensor's measurement origin coordinate system [7]. In the robotic manufacturing system, hand-eye calibration needs to be accurate, convenient, and efficient. In this paper, from the perspective of 3D reconstruction, the ultimate purpose of hand-eye calibration is to make the reconstruction result more accurate.

Therefore, the method of this paper starts from the reconstruction perspective, and first analyzes the absolute and relative errors of the 3D reconstruction points caused by the hand-eye calibration errors. In order to reduce the reconstruction error, a translation measurement strategy is proposed in this paper. On the one hand, this measurement method can reduce the reconstruction error, and on the other hand, it can accurately calibrate the rotation component of the hand-eye calibration matrix. In the calibration method, the error-index based on the 3D reconstructed point clouds is proposed for optimization, and the prediction of the Gaussian process is added to the traditional particle swarm optimization method to accelerate the convergence. After completing the calibration of the rotation component, the multi-pose scanning method is used to calibrate the translation component. Finally, the proposed method is compared with the traditional methods.

To improve the readability, the main contribution is presented as follows:

- (1) The relationship between the hand-eye calibration error and the 3D reconstruction error is analyzed, and based on it, a calibration method is proposed.
- (2) When calibrating the rotation component of the hand-eye relationship, the influence of the line laser sensor measurement error is similarly avoided.
- (3) The authors propose a hand-eye calibration error-index from the perspective of 3D reconstruction so that the calibration directly affects the measurement result instead of the solution accuracy of the hand-eye equation.

The content of this article is organized as follows. Section 2 introduce related works. The research method is described in Section 3, including the theoretical analysis of errors and calibration methods for the rotation component and the translation component. The results are in Section 4, including simulation experiments and real experiments, and the discussion and conclusion are in Sections 5 and 6.

2. Related works

At present, the methods of hand-eye calibration in the “eye-in-hand” system mainly include traditional methods [8–13], optimization methods [14–17], and some probabilistic methods [18,19], and most of the methods are toward visual sensors. In most researches, the robot measures the fixed calibration object in different poses, and then the hand-eye calibration equation $AX = XB$ is constructed. These methods mainly focus on the accuracy of the hand-eye matrix X , and the used evaluation index is usually the relative accuracy of X [20,21].

There are some researches on the hand-eye calibration methods of line laser sensors. Among them, some of the research ideas are similar to the traditional methods, in which the fixed calibration objects are measured at multiple positions to construct a solution equation. A typical way of calibration is to use a standard sphere [5,6,22,23]. In addition, there are some forms of calibration objects with specific shapes, such as dish [24], X-type [25], and multi-step object [26]. Another part of the current research use a special calibration object to construct constraint equations based on the characteristics of the calibration object and solves it through a large amount of measurement data. Paper [27] proposes a calibration method based on a three orthogonal plane, and a single plane calibration method is proposed in [28] to achieve more convenient and robust calibration.

However, from the perspective of three-dimensional reconstruction, these methods currently have three main problems. (1) The evaluation index after calibration is not intuitive enough. Some methods still use relative indicators. Although paper [28] uses the reconstruction index, it only evaluates the flatness after reconstruction, which is not intuitive enough. It is worth mentioning that the paper [29] uses a 3D reconstruction method to achieve an eye-in-hand calibration of lidar sensor through ICP, but this method is difficult to directly apply to the calibration of line laser sensors because the line laser sensors do not have the enough scanning range. (2) Few studies have considered the measurement accuracy of line laser sensors. Taking the Micro-epsilon product scanCONTROL 29X0 as an example, the absolute error of the measurement is $\pm 0.16\%$ [30], that is, at a measurement distance of 300 mm, it has a measurement error of ± 0.5 mm and this part of the error will affect the calibration result. (3) Relatively, some methods have higher manufacturing costs of calibration objects and require complex operations for calibration, which are not suitable for civil industry.

Therefore, in consideration of the above problems, the method of this paper starts from the reconstruction perspective, aiming to propose an intuitive evaluation index and achieve an accurate hand eye calibration based on it.

3. Method

3.1. Error analysis of reconstruction

3.1.1. Absolute position error analysis of reconstruction points caused by hand-eye relationship error

For an arbitrarily shaped object, when a point cloud (set) is as the result of reconstruction after robot scanning, we define this 3D point set as $\{p_i\}$, $i = 1, 2, 3, \dots, N$, where N is the number of points. p_k in the point set $\{p_i\}$ is the coordinate value in Cartesian coordinate system

$$p_k = \begin{bmatrix} p_{kx} \\ p_{ky} \\ p_{kz} \end{bmatrix}. \quad (1)$$

We consider any two points p_1 and p_2 in $\{p_i\}$, and then analyze the effect of the hand-eye calibration matrix error on the change in the vector $v_{12} = p_1 - p_2$ between points. This error represents the change in length and orientation of any part of the point cloud.

First, specify the symbols. We use the common symbol representation method of hand-eye calibration research. A_i is the pose transformation matrix of the robot end (the center of the flange) relative to the robot base at the i th measurement. X is the hand-eye transformation matrix, that is, the pose transformation matrix of the laser sensor measurement origin relative to the robot end. $A_i, X \in SE(3)$. s_i is the measurement result of the laser sensor, which is the Cartesian coordinate of the measured position relative to the origin of the sensor at the i th measurement.

For the two points p_1 and p_2 mentioned above, the results of sensor scanning measurement are s_1 and s_2 .

For real values, it has the following relationship

$$\begin{bmatrix} p_1 \\ 1 \end{bmatrix} = A_1 X_T \begin{bmatrix} s_1 \\ 1 \end{bmatrix}, \quad \begin{bmatrix} p_2 \\ 1 \end{bmatrix} = A_2 X_T \begin{bmatrix} s_2 \\ 1 \end{bmatrix}, \quad (2)$$

where X_T is the real value of the hand-eye transformation matrix.

For measured values calculated with inaccurate (uncalibrated) hand-eye matrix, it has

$$\begin{bmatrix} p_1^M \\ 1 \end{bmatrix} = A_1 X_S \begin{bmatrix} s_1 + \Delta s \\ 1 \end{bmatrix}, \quad \begin{bmatrix} p_2^M \\ 1 \end{bmatrix} = A_2 X_S \begin{bmatrix} s_2 + \Delta s \\ 1 \end{bmatrix}, \quad (3)$$

where X_S is the setting value of the hand-eye matrix for calculation which is inaccurate and needed to be calibrated. p_1^M and p_2^M are the two reconstructed point positions calculated from the set value X_S . It is noted that A_1 and A_2 are the end poses of the robot and they are also considered to be relatively accurate (better than 0.1 mm) after the robot kinematics calibration.

Among the above quantities, s_1 and s_2 are the measured values from the line laser sensor. What needs to be considered is that there is an absolute error of all measured points of one line in this measurement result, and the accuracy is directly proportional to the measurement value. However, when we measure a uniform material and flat surface within a small measuring height changes (smaller than 50 mm), the precision of the measured value is high, which means the relative errors of the points in different lines are relatively small (better than 0.1 mm). Therefore, in this paper, when we scan the configured calibration object, the error of different measurement points is approximately set to Δs . This content has been verified in Section 4.2.2 of this paper.

Then, it can get from Eq. (2) to

$$\begin{bmatrix} v_{12} \\ 0 \end{bmatrix} = A_1 X_T \begin{bmatrix} s_1 \\ 1 \end{bmatrix} - A_2 X_T \begin{bmatrix} s_2 \\ 1 \end{bmatrix}, \quad (4)$$

Since $A_i, X \in SE(3)$, they can be split into rotation matrix R and translation matrix t . So it can be got as

$$\begin{bmatrix} v_{12} \\ 0 \end{bmatrix} = \begin{bmatrix} R_{A_1} & t_{A_1} \\ 0 & 1 \end{bmatrix} \begin{bmatrix} R_{X_T} & t_{X_T} \\ 0 & 1 \end{bmatrix} \begin{bmatrix} s_1 \\ 1 \end{bmatrix} - \begin{bmatrix} R_{A_2} & t_{A_2} \\ 0 & 1 \end{bmatrix} \begin{bmatrix} R_{X_T} & t_{X_T} \\ 0 & 1 \end{bmatrix} \begin{bmatrix} s_2 \\ 1 \end{bmatrix}. \quad (5)$$

Expanding the right formula of Eq. (5), it can be obtained as

$$\begin{aligned} v_{12} &= V_1 + V_2 + V_3, \\ V_1 &= R_{A_1} R_{X_T} s_1 - R_{A_2} R_{X_T} s_2, \\ V_2 &= (R_{A_1} - R_{A_2}) t_{X_T}, \\ V_3 &= t_{A_1} - t_{A_2}. \end{aligned} \quad (6)$$

Similarly, for Eqs. (3), we get

$$\begin{aligned} v_{12}^M &= V'_1 + V'_2 + V'_3, \\ V'_1 &= R_{A_1} R_{X_S} s_1 - R_{A_2} R_{X_S} s_2 + (R_{A_1} - R_{A_2}) R_{X_S} \Delta s, \\ V'_2 &= (R_{A_1} - R_{A_2}) t_{X_S}, \\ V'_3 &= t_{A_1} - t_{A_2}. \end{aligned} \quad (7)$$

Eq. (6) is subtracted from Eq. (7) to get

$$\begin{aligned} Err &= R_{A_1} (R_{X_T} - R_{X_S}) s_1 - R_{A_2} (R_{X_T} - R_{X_S}) s_2 \\ &\quad + (R_{A_1} - R_{A_2}) (t_{X_T} - t_{X_S} - R_{X_S} \Delta s), \end{aligned} \quad (8)$$

where $Err = v_{12} - v_{12}^M$ represents the error of vector between two points in the object caused by the error of the hand-eye matrix. This item more intuitively evaluates the effect of the hand-eye matrix on the accuracy of 3D reconstruction. The ideal calibration result will make this term approach to 0, which is the true distortion-free reconstruction.

It can be analyzed from Eq. (8) that when the robot scans, in order to reduce the error of the reconstruction point, a good strategy is to keep $R_{A_1} = R_{A_2}$. At this time, both the translation component error of the hand-eye matrix and the line laser measurement error are eliminated.

3.1.2. Relative position error analysis of reconstruction points caused by hand-eye relationship error

In the previous section, the analysis of is toward the absolute error in the robot base coordinate system. However, in actual measurement, in most cases, it is not necessary to obtain the absolute position of the reconstructed point clouds and only the precise point cloud shape is needed. Meanwhile, it is difficult to obtain the true value of the

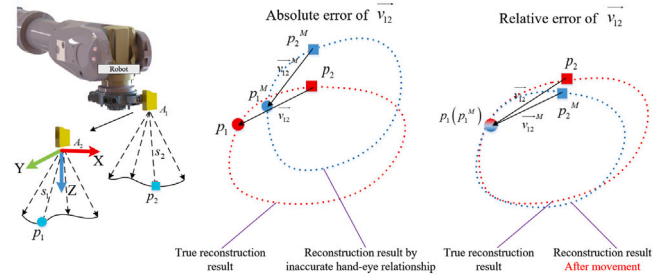


Fig. 1. Schematic diagram of absolute and relative errors.

absolute position. So the absolute error is difficult to be measured and calculated. Therefore, the analysis of the relative errors is necessary.

The relative error is defined as the error between the real point cloud and the measured point cloud after a proper rigid transformation T , $T \in SE(3)$, and this transformation T should minimize the relative error as much as possible. The absolute and relative errors are shown in Fig. 1 and the definition of the coordinate axis of the laser sensor is shown in the figure. The light plane measured by the sensor is the X-Z plane, and the Y-axis is perpendicular to this plane.

In this section, it is analyzed whether there are other possible wrong hand-eye calibration values that make the relative error 0 under the set calibration conditions, which proves that accurate reconstruction and accurate calibration are equivalent.

Suppose after the transformation T , p_1 coincides with p_1^M . Firstly consider the sensor measurement error Δs , when $R_{A_1} = R_{A_2}$, that is, the end orientation does not change during the measurement, the point cloud can be first translated by T_1 , $T_1 \in SE(3)$ to eliminate this part of the error.

$$\begin{aligned} A_1 X_S \begin{bmatrix} s_1 \\ 1 \end{bmatrix} &= T_1 A_1 X_S \begin{bmatrix} s_1 + \Delta s \\ 1 \end{bmatrix}, \\ A_2 X_S \begin{bmatrix} s_2 \\ 1 \end{bmatrix} &= T_1 A_2 X_S \begin{bmatrix} s_2 + \Delta s \\ 1 \end{bmatrix}, \\ T_1 &= \begin{bmatrix} E_3 & -R_A R_{X_S} \Delta s \\ 0_{1 \times 3} & 1 \end{bmatrix}, R_A = R_{A_1} = R_{A_2}. \end{aligned} \quad (9)$$

Therefore, $T = T_2 T_1$, which represents p_1 coincides with p_1^M after two sequential movements T_1 and T_2 .

$$\begin{bmatrix} p_1 \\ 1 \end{bmatrix} = T_2 T_1 \begin{bmatrix} p_1^M \\ 1 \end{bmatrix}, \quad (10)$$

$$A_1 X_T \begin{bmatrix} s_1 \\ 1 \end{bmatrix} = T_2 T_1 A_1 X_S \begin{bmatrix} s_1 + \Delta s \\ 1 \end{bmatrix} = T_2 A_1 X_S \begin{bmatrix} s_1 \\ 1 \end{bmatrix}. \quad (11)$$

Since p_1 is the measurement when robot is in pose A_1 . It can also be found p_2, p_3, p_4 in robot pose A_2, A_3 , and A_4 . Meanwhile, since A_2, A_3 , and A_4 are parallel to A_1 , it can be obtained that $A_2 = A_1 \begin{bmatrix} E_3 & \Delta t_{12} \\ 0 & 1 \end{bmatrix}$. Similarly, it has Δt_{13} and Δt_{14} for A_3 and A_4 .

Then, take p_2 as an example, it can also get the equation like Eq. (11).

$$A_2 X_T \begin{bmatrix} s_2 \\ 1 \end{bmatrix} = T_2 A_2 X_S \begin{bmatrix} s_2 \\ 1 \end{bmatrix}. \quad (12)$$

Since A_2 and A_1 have the translation relationship, it can be found that

$$A_1 \begin{bmatrix} E_3 & \Delta t_{12} \\ 0 & 1 \end{bmatrix} X_T \begin{bmatrix} s_2 \\ 1 \end{bmatrix} = T_2 A_1 \begin{bmatrix} E_3 & \Delta t_{12} \\ 0 & 1 \end{bmatrix} X_S \begin{bmatrix} s_2 \\ 1 \end{bmatrix}. \quad (13)$$

Meanwhile, since $X_S, X_T \in SE(3)$, Eq. (13) can be derived as

$$T_2 A_1 X_T \begin{bmatrix} s_2 + R_T^{-1} \Delta t_{12} \\ 1 \end{bmatrix} = A_1 X_S \begin{bmatrix} s_2 + R_S^{-1} \Delta t_{12} \\ 1 \end{bmatrix}. \quad (14)$$

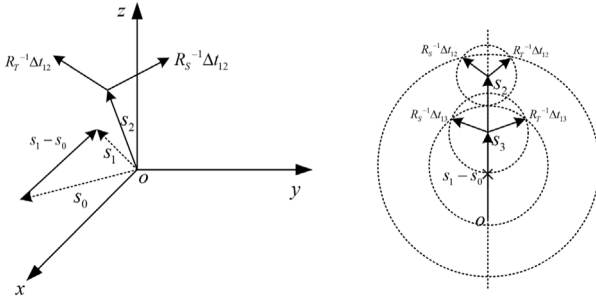


Fig. 2. Schematic diagram of the transformation.

Then, we can get another point p_0 when the robot is in A_1 , and for p_0, p_1, p_2, p_3, p_4 , Eq. (14) is applied. Define $W = X_S^{-1} A_1^{-1} T_2 A_1 X_T \in SE(3)$, it can be obtained that the transformation W satisfy the five points, as shown in Eq. (15).

$$W \begin{bmatrix} s_0 & s_1 & s_2 + R_T^{-1} \Delta t_{12} & s_3 + R_T^{-1} \Delta t_{13} & s_4 + R_T^{-1} \Delta t_{14} \\ 1 & 1 & 1 & 1 & 1 \end{bmatrix} = \begin{bmatrix} s_0 & s_1 & s_2 + R_S^{-1} \Delta t_{12} & s_3 + R_S^{-1} \Delta t_{13} & s_4 + R_S^{-1} \Delta t_{14} \\ 1 & 1 & 1 & 1 & 1 \end{bmatrix}. \quad (15)$$

Therefore, the analysis in this section becomes to prove that if and only when $R_S = R_T$, Eq. (15) holds, that is, the relative error is zero. Two cases are analyzed as below.

The first case is that there exists another point s_5 when the robot is in A_1 , and s_5 is not collinear with s_0, s_1 , then W can be uniquely calculated by $W \begin{bmatrix} s_0 & s_1 & s_5 \\ 1 & 1 & 1 \end{bmatrix} = \begin{bmatrix} s_0 & s_1 & s_5 \\ 1 & 1 & 1 \end{bmatrix}$, that is $W = E_4$, which represents not move. Therefore, for the last three points s_2, s_3 and s_4 , only when (1) the axis of the transformation (expressed in the form of axis-angle, the same below) $R_S R_T^{-1}$ is in the same direction as $\Delta t_{12}, \Delta t_{13}$ and Δt_{14} , or (2) $R_S = R_T$, it has $W = E_4$. However, when the robot is moving along a polyline, that is $\Delta t_{12} \neq \Delta t_{13}$, therefore, only $R_S = R_T$ can satisfy the situation that the relative error of the reconstructed point cloud is zero. However, in this case, a point s_5 is needed, which demands that one scanning result must be a polyline. Therefore, it requires that the scanned object must not be a plane. In actual calibration application, compared with other types of calibration object, the plane calibration object has the advantages of easy processing, low cost, and simple measurement. So the second case should be analyzed.

In the second case, there only has a straight line in one scanning result, therefore, W cannot calculated directly without s_5 . Expand Eq. (15) and perform some subtractions on several items in the matrix. It can be get

$$\begin{aligned} R_W (s_0 - s_1) &= s_0 - s_1, \\ R_W (s_2 + R_T^{-1} \Delta t_{12} - s_1) &= s_2 + R_S^{-1} \Delta t_{12} - s_1, \\ R_W (s_3 + R_T^{-1} \Delta t_{13} - s_1) &= s_3 + R_S^{-1} \Delta t_{13} - s_1, \\ R_W (s_4 + R_T^{-1} \Delta t_{14} - s_1) &= s_4 + R_S^{-1} \Delta t_{14} - s_1. \end{aligned} \quad (16)$$

For the above formula, it can be understood from the geometric meaning that the four vectors on the left side of the formula are transformed to the four vectors on the right side by a rotation.

Since the vector $s_0 - s_1$ do not change after the rotation, the axis of R_W must be along $s_0 - s_1$, as shown in Fig. 2. For s_2 , the two points before and after the rotation ($s_2 + R_T^{-1} \Delta t_{12}, s_2 + R_S^{-1} \Delta t_{12}$) is shown in the left part of the figure. For $R_T^{-1} \Delta t_{12}$, only two situations can satisfy Eq. (16). The first is that $R_S = R_T$, and $W = E_4$. The second is shown in the right part of Fig. 2, in which $s_0 - s_1$ is perpendicular to this visible paper. There have two conditions. (1) The distances from two points to the line $s_0 - s_1$ should be equal, which is the outer circle of dotted line in the figure. (2) The length of $R_T^{-1} \Delta t_{12}$ and $R_S^{-1} \Delta t_{12}$ are equal, which is the inner circle of dotted line. Therefore, only one possibility

can meet the two conditions, which is shown in the figure. Similarly, for s_3 , it is shown in the figure. When the directions of Δt_{12} and Δt_{13} are different, for $\Delta R = R_S^{-1} R_T$, the unique value can be solved. Therefore, when the direction of Δt_{14} is not coplanar with Δt_{12} and Δt_{13} , we cannot find a $\Delta R = R_S^{-1} R_T$ which satisfies Eq. (15). Therefore, the second situation is invalid, and only the first situation can achieve an accurate reconstruction considering the relative error.

In summary, when a robot and a line laser sensor are used to measure and perform three-dimensional reconstruction of an object, it can be concluded that when the robot keeps the same orientation during scanning and moves as a three-segment polyline which is not in one plane, the only case when the relative error is 0 is that the set value of the rotation matrix part in the hand-eye matrix is equal to the true value ($R_S = R_T$). Therefore, this measurement strategy can be used in a calibration method. When the relative error of the measured point cloud is near 0, the rotation component calibration of the hand-eye matrix is achieved.

At the same time, it can be concluded that if only considering the shape accuracy of the scanned object, this translation measurement strategy can achieve an accurate measurement only after the rotation component calibration.

3.2. Calibration method for rotation component

In the calibration, since the ultimate goal is to make the error near 0, an optimization method is used. Meanwhile, the purpose of the optimization is to make the shape of the reconstructed point cloud closer to the standard point cloud in order to achieve more accurate measurements.

However, there are two main problems in the actual calibration and measurement process. (1) the result point cloud of reconstruction is from multiple line-scanning while the standard point cloud (seen as the true value) is sampled from a CAD (Computer Aided Design) model, so the correspondence between the points of the two point clouds is unknown, that is, it is difficult to directly find the corresponding point p_i^M of p_i . Meanwhile, the movement T of the point cloud in Section 3.1.2 is unknown. (2) In actual scanning, the result often involves many other parts rather than only the calibration object itself, such as the plane on which the calibration object is placed, the side part of the object and so on, so it needs to remove through processing algorithms. However, it is difficult to have a universal method to achieve the desired removing effect. Therefore, this will reduce the efficiency of the calibration.

3.2.1. Determination of optimization objective for calibration

In order to solve the first problem, a point cloud registration method based on Iterative Closest Point (ICP) [31] is applied in this paper. The main idea of ICP is to minimize the distance between the nearest points, and its cost function is

$$E(R, t) = \frac{1}{n} \sum_{i=1}^n \|q_i - (R p_i + t)\|^2, \quad (17)$$

where p_i is a point in the point cloud reconstructed according to the current setting value of the hand-eye matrix. q_i is the point closest to p_i in the standard point cloud. $E(R, t)$ is the mean square error of the distances between all points in the reconstructed point cloud and their nearest points in the standard point cloud. The ICP algorithm is to calculate the best R and T by minimizing $E(R, t)$, that is

$$R_0, t_0 = \operatorname{argmin} (E(R, t)), \quad (18)$$

where R_0 and t_0 is the output value of ICP algorithm when $E(R, t)$ is minimum, and set $E_0 = \min E(R, t)$. Therefore, in the first problem of Section 3.2, when the standard point cloud is set densely enough, the point-pair of nearest points output by the ICP algorithm can be used as the corresponding point-pair. Although it is not a true corresponding point-pair, the trend of the point-pair's distance error is the same as

the real. When the error is near 0, it can also indicate that the reconstruction result is correct, and the calibration is completed. Meanwhile, T can get from ICP calculation result.

So, for each given R_S , the minimum error E_0 calculated by ICP algorithm can be set as the objective function of the calibration algorithm. In order to make the results more intuitive, we use the root mean square error $E_{rmse}(R_S) = \sqrt{E_0(R_S)}$.

$$E_{rmse}(R_S) = \min_{R, t} \left(\sqrt{\frac{1}{n} \sum_{i=1}^n \|q_i - (Rp_i(R_S) + t)\|^2} \right), \quad (19)$$

$$p_i(R_S) = AX_S s_i, i = 1, \dots, n,$$

$$X_S(R_S) = \begin{bmatrix} R_S & 0_{3 \times 1} \\ 0 & 1 \end{bmatrix}.$$

From another perspective, the optimized cost function directly reflects the reconstruction accuracy of the scanning results, which makes the calibration directly affect the scanning effect. Therefore, this optimized result is more intuitive and can be used as an error-index to measure the accuracy of linear laser sensor hand-eye calibration.

Therefore, the ICP method solves the first problem in Section 3.2. However, it should be emphasized that using the ICP method has two disadvantages. (1) The current optimization method is a double layer optimization. For each generation of $E_{rmse}(R_S)$ calculation, an ICP optimization calculation must be performed once, which will cause the overall calculation slow and reduce the calibration efficiency. (2) When there is no good initialization, the algorithm easily enters the local optimum and generates incorrect calculation values. Therefore, an improved particle swarm algorithm method is adopted to solve the first problem, which will be described in detail in Section 3.2.2. At the same time, the selected part of the point cloud used for ICP is also simplified as much as possible in order to increase the calculation speed. For the second problem, since the laser sensor and the connection block have nominal designed sizes as the initial value, which are close to the true values, a correct optimization can be ensured, and the final experimental results also confirm this.

3.2.2. Optimization method for calibration based on particle swarm optimization - Gaussian process(PSO-GP) method

In this section, in order to avoid the local optimization, a particle swarm algorithm-based method is adopted. The reasons for using particle swarm optimization are listed below. (1) It is difficult to find the derivative of the objective function because the function involves a result of ICP optimization. (2) The enumeration method is too computationally intensive. (3) The branch-and-bound method is useful in the global registration of point clouds [32]. However, due to the deformation and movement of the point cloud, it is difficult to find a suitable lower bound. Therefore, an optimization method based on particle swarm optimization is used. At the same time, considering the problem of excessive calculation amount in the optimization function, the optimal global prediction is performed by each generation in the particle swarm algorithm using the Gaussian process to accelerate the convergence speed.

(1) Determination of variables for the optimization method

As can be seen from Section 3.1, through optimization (calibration) of the R_X matrix, accurate 3D reconstruction can be achieved. As a rotation matrix, R_X has only three degrees of freedom, so this paper uses Euler angles with the Z-Y-X rotation order as the variable. It should be noted that Euler angles do not correspond one-to-one with the rotation matrix, which leads to optimization failure. That is, multiple sets of Euler angles can represent the same rotation matrix. However, in the case of non-universal joint deadlock (that is, the p angle is not 90 or -90 degrees), for one R_X , each angle has a unique solution within positive and negative 180 degrees. In actual calibration, since the initial value is got from the designed value, the true value is usually not much different from the design value (which is far less than

180 degrees). So unless the initial value is near (0, $\pm 90, 0$), Euler angles can be applied to this optimization method. If the initial value is really near it, adding a virtual joint to the end of the robot's kinematic model is needed, and it performs a posture transformation to make the initial value far from (0, $\pm 90, 0$).

(2) Particle Swarm Optimization

The optimization of the particle swarm algorithm [33] is realized by stepwise iteration. First, particle swarm is initialized randomly. In each subsequent iterative update process, all particles will find two optimal solutions to fly. The first is their optimal historical position ($pBest$, regarded as their own experience), and the second is the optimal global position of all particles in the particle swarm ($gBest$, regarded as peer experience). According to specific rules, they update the speed and position, and gradually find the optimal value. The particles are defined as follows.

$$D_{[i]} \equiv \begin{Bmatrix} \mathbf{x}_{[i]} = [r_{[i]}, p_{[i]}, y_{[i]}] \\ \mathbf{v}_{[i]} = [v_{r_{[i]}}, v_{p_{[i]}}, v_{y_{[i]}}] \end{Bmatrix}, \quad (20)$$

where r, p, y are the Euler angle value of the R_S . The particles are updated as follows.

$$\begin{aligned} \mathbf{v}_i^{k+1} &= \omega \mathbf{v}_i^k + c_1 \text{Rand}_1(\mathbf{B}_{hi}^k - \mathbf{x}_i^k) + c_2 \text{Rand}_2(\mathbf{B}_g^k - \mathbf{x}_i^k), \\ \mathbf{x}_i^{k+1} &= \mathbf{x}_i^k + r \mathbf{v}_i^{k+1}, \end{aligned} \quad (21)$$

where $\mathbf{B}_{hi}$ is the optimal historical position of the particle itself, $\mathbf{B}_g$ is the historically optimal position of the particle group, and k and $k+1$ represent the current generation and the next generation of the particle. c_1 and c_2 are the weights of the particle's optimal value and the optimal global value. Rand_1 and Rand_2 are random values in the range of 0 to 1. ω controls the influence of the original speed on the current speed. r is a constraint factor that controls the effect of the update speed on the update position.

(3) Gaussian Process

In this paper, the optimal global value of particles in each generation will be predicted by the Gaussian process [34], thereby directing the particle swarm to the optimal region.

a. The construction method of Gaussian process

The Gaussian process is an extension of the multivariate Gaussian distribution, and any finite number of random variables obey it. Gaussian processes can be constructed from mean functions m_p and covariance functions k_p .

$$g_p(x) \sim GP(m_p(x), k_p(x, x')), \quad x, x' \in X. \quad (22)$$

b. Training of Gaussian process

In the training of Gaussian processes, the parameters of the prior distribution are called hyperparameters.

$$\theta = [p; \sigma_\epsilon], \quad (23)$$

where p is the construction parameter in the mean function and covariance function, and σ_ϵ is the variance of the noise distribution.

Maximum Likelihood Method is commonly used to determine hyperparameters

$$\theta_* = \arg \min L(\theta), \quad (24)$$

where

$$\begin{aligned} L(\theta) &= -\log p(\hat{y} = y|x) \\ &= -\frac{1}{2} (y - m_p(x))^T (K_p(x, x) + \sigma_\epsilon^2 I)^{-1} (y - m_p(x)) \\ &= -\frac{1}{2} \log |K_p(x, x) + \sigma_\epsilon^2 I| - \frac{n}{2} \log 2\pi. \end{aligned} \quad (25)$$

c. Prediction of Gaussian process

After training is complete, the prior probability distribution of the dependent variable can be obtained. To the prediction points $(x_*, \hat{y}_*)$, the new vector composed of two sets of random variables still

Algorithm 1 Optimization Method based on PSO-GP

Input: $r_0, p_0, y_0, bound$
Output: optimal result r_*, p_*, y_*, E_{rmse}
Initialize particle swarm and parameters c, r, ω .
Compute initial value of E_{rmse0} for particle r_0, p_0, y_0 .
Create initial particles $\{r_i\}^1, \{p_i\}^1, \{y_i\}^1$ in given parameter range $bound$ and calculate $\{E_{rmsei}\}^1$.
Select individual optimal value B_{hi}^1 and group optimal value B_g^1 .
Set $k = 1$ (generation)
while Convergence condition not satisfied **do**
Set $k = k + 1$
Calculate the predicted global optimum using all generations particles by Gaussian process
 $(x_{GP}^{k-1}, GP_{rmse}^{k-1}) = GP(\{r_i\}^{1,...,k-1}, \{p_i\}^{1,...,k-1}, \{y_i\}^{1,...,k-1})$.
Compute $E_{x_{GP}^{k-1}}$ of predicted best particle x_{GP}^{k-1} .
if $E_{rmse}(x_{GP}^{k-1}) < E_{rmse}(B_g^{k-1})$ **then**
 $B_g^{k-1} = x_{GP}^{k-1}$
end if
Update position and velocity vectors for each particle $\{r_i\}^k, \{p_i\}^k, \{y_i\}^k$.
Compute $\{E_{rmsei}\}^k$ of every particle.
Update individual optimal value B_{hi}^k and group optimal value B_g^k .
end while
The optimal value E_{rmse} of B_g^k and its corresponding particle r_*, p_*, y_* are the final output.

obeys the joint prior probability distribution. the posterior probability distribution of the random variable $\hat{y}_*$ is

$$\hat{y}_* | x_*, p, \sigma_e, x, \hat{y}_{=y} \sim \mathcal{N}(\mu, \sigma), \quad (26)$$

where

$$\begin{aligned} \mu &= m_p(x_*) + k_p(x_*, x) (K_p(x, x) + \sigma_e^2 I)^{-1} (y - m_p(x)), \\ \sigma &= (k_p(x_*, x_*) + \sigma_e^2 I) - k_p(x_*, x) (K_p(x, x) + \sigma_e^2 I)^{-1} k_p(x, x_*). \end{aligned} \quad (27)$$

(4) PSO-GP optimization Algorithm for calibration

The PSO-GP optimization method is shown in Algorithm 1. The key is that in the calculation of each generation of particles, the calculated values of all particles in the history are used as samples, and the optimal global prediction is performed using the Gaussian process. Then, the predicted optimal particle is used to calculate the objective function again. If the value is smaller than the current historical optimal value, the optimal historical value is updated.

In our algorithm, since we focus on the accuracy of the calibration, the convergence condition is set to $E_{rmse}(B_g^k) < E_{Threshold}$ or $E_{rmse}(B_g^k) - E_{rmse}(B_g^{k-1}) < B_{Threshold}$ in successive n times of generations.

3.3. Calibration method for translation component

In the previous chapters, we have proved that when using the translation strategy for robot scanning, it only needs to calibrate the rotation component of the hand-eye matrix to complete an accurate 3D reconstruction. Therefore, the calibration method for the translation component is not the focus of this article. However, in some measurements, due to the characteristics of the target, the robot needs to adjust the orientation. Therefore, this paper proposes a calibration method for the translation component after the rotation component is calibrated.

In this section, we use the method of constructing the $AX = XB$ equation to obtain the calibration results. During the calibration, the robot scans the calibration object fixed at a stable position multiple

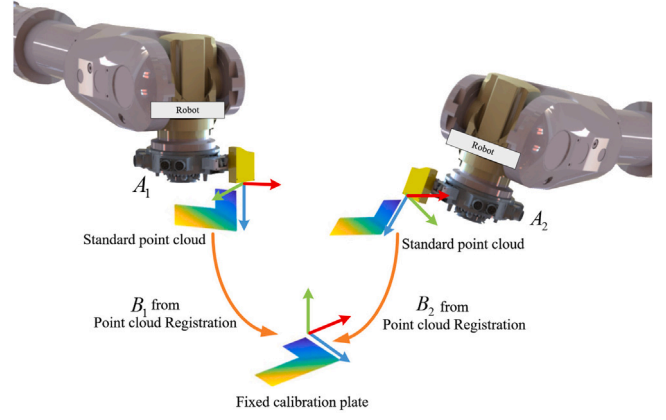


Fig. 3. Calibration method of translation component.

times from different poses A_i , and it obtains the measured pose B_i of the calibration object relative to the line laser sensor, and then constructs the equation $AX = XB$ to solve hand-eye relationship matrix X . A and B in this equation are relative values of measured value $\{A_i\}$ and $\{B_i\}$.

It should be noted that the line laser sensor is different from the traditional vision sensor because only one-dimensional data of the calibration object can be obtained in once measurement. Therefore, if we want to obtain a complete calibration object measurement, a set of parallel movements by the robot is needed to get the corresponding measurements to combine and reconstruct the object. Since the orientation of the sensor has been calibrated, the movement direction of the robot scanning can be obtained, which is along the Y-axis of the laser coordinate system. It can be calculated by

$$\begin{bmatrix} dir \\ 0 \end{bmatrix} = A_i \begin{bmatrix} R_X & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 0 \\ 1 \end{bmatrix} - \begin{bmatrix} t_{A_i} \\ 1 \end{bmatrix}. \quad (28)$$

For the i th measurement, when performing intermittent translation $\{A_j\}$ in this direction, accurate 3D reconstruction can be completed. In the reconstructed point cloud $\{x_j, y_j, z_j\}$, the x_j and z_j values of each scan result are the sensor measurement values x and z , and the y_j value is the distance between the position of the robot at the j th point and the starting point A_1 of scanning. Therefore, the results of all this series of scans are combined as the i th set of measurement result. After the reconstruction of the point cloud, the pose B_i of the point cloud relative to the laser coordinate system can be obtained by point cloud ICP registration between scanned point clouds and the standard point cloud, as shown in Fig. 3. In addition, it should be noted that due to the point cloud registration method, the calibration object needs to be an asymmetric pattern to prevent multiple solutions of the registration. In this article, an L-shaped calibration block is used.

After the data is acquired, it can be used to solve the hand-eye equation. After expanding the $AX = XB$ equation according to the rotation component and translation component, it can be derived that

$$(R_A - I) t_X = R_X t_B - t_A. \quad (29)$$

For multiple sets of data, the translation component t_X can be calculated by using the least squares method.

4. Result

4.1. Simulation experiment for calibration

4.1.1. Simulation data generation method

At the beginning of the simulation, the scanning simulation data generation is first performed. Before data generation, the data of the

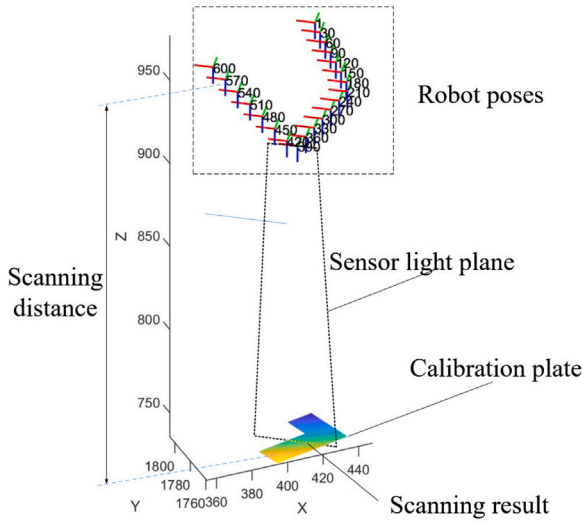


Fig. 4. Configuration of simulation data generation.

workpiece to be scanned is created. Since the model is L-shaped, there are six key points $\{keyP_j\}$ and six line segments $\{keyL_j\}$ connected end to end which is expressed as

$$\frac{x - x_{keyP_j}}{x_{keyP_{j+1}} - x_{keyP_j}} = \frac{y - y_{keyP_j}}{y_{keyP_{j+1}} - y_{keyP_j}} = \frac{z - z_{keyP_j}}{z_{keyP_{j+1}} - z_{keyP_j}}. \quad (30)$$

On the other hand, to determine the scanning poses of the robot, it can be known from Section 3.1 that the scanning points must not be collinear. Therefore, dense and equidistant sampling is performed on the polyline, and at the same time, the appropriate pose is given at each sampling point. Generally, the more perpendicular the light plane of the laser sensor to the plane of the object to be measured, the higher the measurement accuracy is. The scanning configuration is shown in Fig. 4. After generating the robot poses and calibration plate data, the data measured in each pose can be easily calculated through coordinate transformation.

$$\begin{bmatrix} keyP_j^i \\ 1 \end{bmatrix} = A_i \begin{bmatrix} keyP_j \\ 1 \end{bmatrix}. \quad (31)$$

Coordinate transformation is performed for each measurement, $keyP_j^i$ is the key points after the i th measurement conversion, and the calculated measurement point on the edge of the calibration plate is the point of the key line segment at $y = 0$.

4.1.2. 3D reconstruction error theory verification

In this part, the true value of the rotation Euler angle is set to $r = 59.1^\circ, p = -0.2^\circ, y = -0.4^\circ$, in order to verify the relative error theory previously proposed, the distribution is reconstructed with different Euler angle values. The reconstruction results are shown in Fig. 5.

It can be analyzed from the figure that when the design value $r = 60^\circ, p = 0^\circ, y = 0^\circ$ is used, which is also the initial value, it can be seen that the model has a certain error in the red dotted line in Figure c (the green and purple parts in Figure c are from Figure a and b, and the result is after registration). As can be seen from the remaining figures, the errors caused by different values of Euler angles are obvious, which also validates the error theory.

4.1.3. Calibration of rotation component

PSO-GP is used for optimization in this part to achieve calibration. The configuration of the optimization algorithm is as follows, the size of the particle is 20, $r = 1, \omega = 0.8, c_1 = 0.5, c_2 = 2, E_{threshold} = 0.001$ mm, $B_{threshold} = 5 \times 10^{-5}$ mm, $n = 10$. The initial value is $r = 60^\circ, p = 0^\circ, y = 0^\circ$ while the bound is from $r = 58^\circ, p = -2^\circ, y = -2^\circ$ to $r = 62^\circ, p = 2^\circ, y = 2^\circ$. When predicting the optimal value of the Gaussian process, the minimum value is selected by predicting the objective function through the enumeration method, and the step size of the selected variable is 0.1° .

The program runs on the processor Intel Core i7-6500U with a central frequency of 2.5 GHz. PSO-GP method and PSO method are used for calculation and comparison, respectively. With the same optimization function and the same algorithm parameter configuration, the PSO method takes a total time of 620.545 s and undergoes a total of 45 iterations. It has undergone 900 calculations of the objective function. The total time of the ICP algorithm is 607.441 s. The PSO-GP method only takes 244.814 s in total and only experienced seven iterations. Among them, 146 calculations of the objective function are performed. The total time of the ICP algorithm is 231.438 s, and the time of the Gaussian process is 16.426 s.

The comparison result of the objective function values of the two methods, and the positions of the optimal particles of each generation are shown in Fig. 6, 7 and 8. Both algorithms converge to the correct solution. It can be seen that the PSO-GP algorithm increases the speed of convergence, from 45 generations of the PSO algorithm to 7 generations with a decrease of 84.4%. The predicted value of GP is inaccurate in the previous generations, and this is because there are too few samples in the previous generations of the Gaussian process. As the sample increases, the predicted value gradually becomes accurate. At the same time, it can be seen that the calculation time is only reduced by 61.9%, which is less than 84.4%, this is because the ICP algorithm

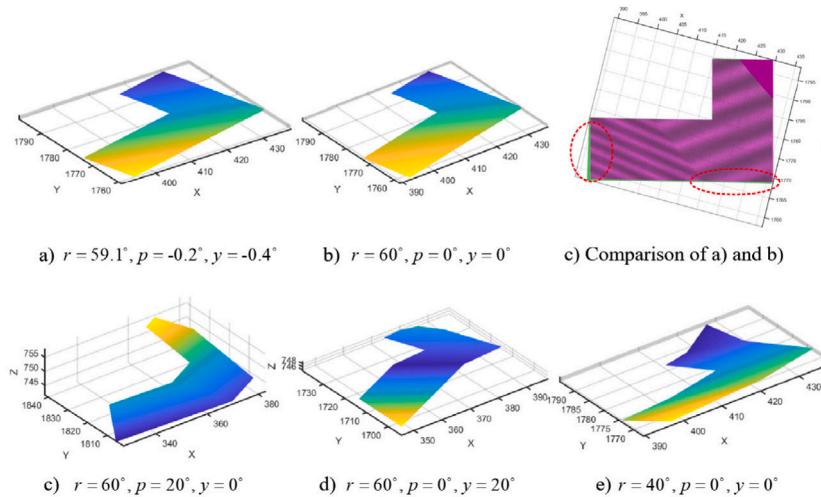


Fig. 5. 3D reconstruction error (unit: mm). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

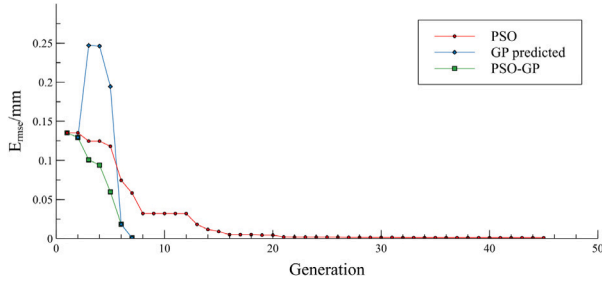


Fig. 6. Comparison of optimization processes.

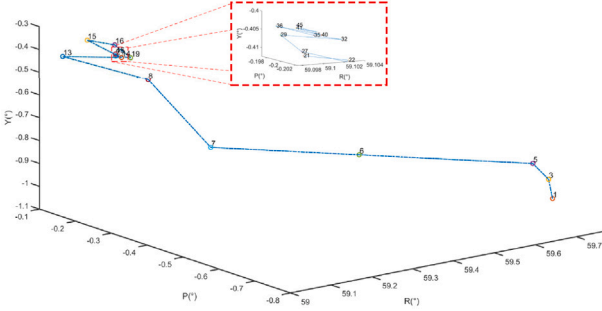


Fig. 7. Optimal particles in each generation of the PSO method.

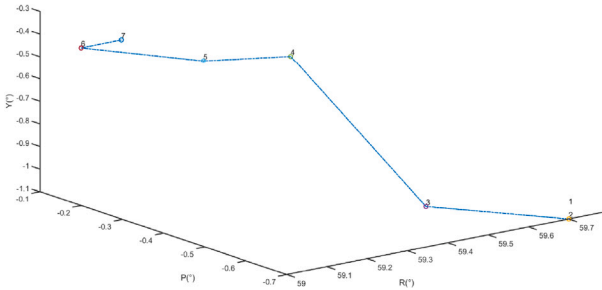


Fig. 8. Optimal particles in each generation of the PSO-GP method.

is an optimization algorithm. Its calculation time is different at each time, the closer the current value is to the real value, the faster the calculation will be.

In order to analyze the impact of measurement noise, noise is added during the generation of measurement data, and the simulation is performed. There are two types of noise, one is the absolute error, and d_s is added to each measurement point. The other is relative error noise, and a random error of $\{d_{si}\}$ is added to each measurement point. The results are shown in Table 1. It can be seen from the results in the table that the absolute noise has no effect on the calibration results, which is consistent with the analysis of the previous content. For relative errors, when the errors are small, they have a small effect on the calibration results, but when the error increases to ± 0.5 mm, it affects the calibration results by about 0.05 degrees, and the reconstruction accuracy is 0.2893 mm.

4.2. Real experiment for calibration

In the real experiment, an ABB IRB6700 robot and a Micro-Epsilon sensor scanCONTROL 2900-100/BL were used for experiments. The pose recording of the robot was implemented through the SOCKET communication. When the robot moved to each scanning pose, the robot transmitted the current posture, and the PC received the robot pose and turned on the sensors to measure, then recorded data through

Table 1

Comparison of optimization results under different noises.

Noise type	Value (mm)	Optimized r, p, y (°)	$E_{r,mse}$ (mm)
Absolute	0.1	59.1, -0.2, -0.4	0.0008
Absolute	1	59.1, -0.2, -0.4	0.0008
Relative	± 0.01	59.1, -0.2, -0.4	0.0055
Relative	± 0.05	59.1, -0.2, -0.4	0.027
Relative	± 0.1	59.1, -0.2, -0.4	0.0569
Relative	± 0.5	59.063, -0.252, -0.425	0.2893

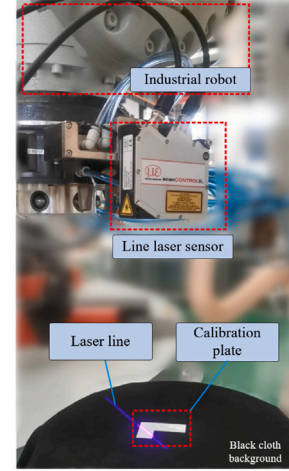


Fig. 9. Experiment platform.

the API interface. After this recording, the command was sent to the robot. Then the robot moves to the next pose and continued the above tasks.

4.2.1. Configuration method of calibration experiment

In order to solve the second problem in Section 3.2 and consider the characteristics of ICP algorithms, this section describes the design and scanning method of the specific calibration plate in the calibration experiment. The specific design is shown in Fig. 9. In order to minimize the work of filtering the point cloud and removing unnecessary areas, the calibration object is designed as a white flat sheet with a thickness of 0.1 mm, and the dimensional accuracy of is less than ± 0.02 mm. When the sensor scans above the calibration plate, the results of the side parts are nearly none so that the complex processing is avoided. A black cloth light-absorbing background is used, when the sensor's exposure time is set to a reasonable value, the measurement of the black cloth part can be almost entirely invalid. Therefore, under this design, the measurement results of the calibration plate can be easily extracted to improve the calibration efficiency.

At the same time, considering the characteristics of the ICP algorithm, the calibration object is designed as an asymmetrical L-shaped plate, and the length scale between the two sides is 3:2. In order to increase the calculation speed, only the edge points in the pattern are used as valid points in point cloud computing. In the scan results of each line, the two ends of the gathered points can be directly obtained as edge points. The number of scan points in one line is 1280, and the distance between each neighbor points is smaller than 0.05 mm. It should be emphasized that in the measurement, the robot moves to dense, discrete poses in turn. Four initial poses can be obtained by simple teaching by human operator, and then all the poses are interpolated by the initial poses. After each arrival, the robot stops, and the sensor measures, which is to obtain a more accurate robot measurement pose record relative to continuous motion.

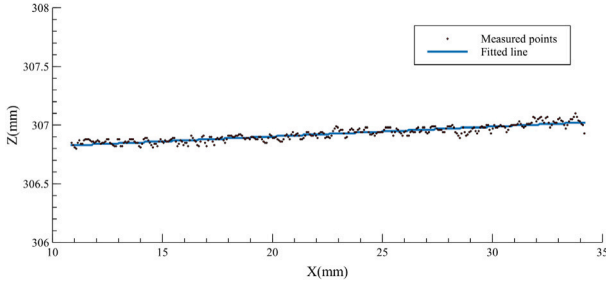


Fig. 10. Measurement results.

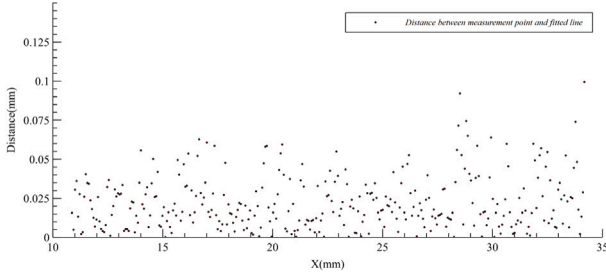


Fig. 11. Distance between measurement point data and fitted line.

4.2.2. Measurement error analysis

In this part, in order to verify the content of the measurement error of the sensor proposed in the theoretical part, one line of the data of the plane measurement of the calibration plate is taken for analysis. The measurement results are shown in Fig. 10. It can be seen that the measured data is very close to a straight line, and the fitted straight line is also shown in the figure. In order to measure the precision of the measurement, the distance between the measurement point and the fitted straight line is shown in Fig. 11. It can be seen that for this straight line measured, the error of most points is less than 0.05 mm. At the same time, in fact, some noise can be removed by filtering algorithms to improve accuracy further. The absolute measurement distance is near 307 mm, and from the overall measurement, the measured distance changes within 50 mm, so according to the sensor index, the absolute error at this time can be calculated to be about 0.46–0.54 mm, which is far greater than the relative error. Therefore, the analytical premise of the theoretical part in Section 3.1 is verified.

4.2.3. Calibration of rotation component

Similar to the simulation part, the trajectory of the robot has 600 dense points on the polyline, and the initial Euler angle value of the rotation component of the hand-eye matrix is $r = 60^\circ, p = 0^\circ, y = 0^\circ$, the objective function is optimized using the PSO-GP optimization method. The result is shown as Fig. 12, from the results it can be seen that when the same configuration was used as the simulation part, the PSO-GP method converged after 17 generations, and the program took 1326.296 s. 356 ICP calculations and 16 Gaussian process calculations were performed. In contrast, the PSO method converged after 29 generations, the program took 2014.02 s, and 580 ICP calculations were performed. The final calculation result is $r = 59.8876^\circ, p = -0.0571^\circ, y = -0.7023^\circ, E_{rmse} = 0.1256$ mm.

The optimization results and initial values were used for 3D reconstruction, as shown in Fig. 13. It can be clearly seen that after calibration, the results with 3D reconstruction are more accurate, especially the distortion of the red dotted line in the figure is almost corrected. From the perspective of reconstruction accuracy, the final reconstruction accuracy can reach 0.1256 mm.

In order to measure the error more intuitively, the error comparison results of all reconstruction points are shown in Fig. 14. It can be

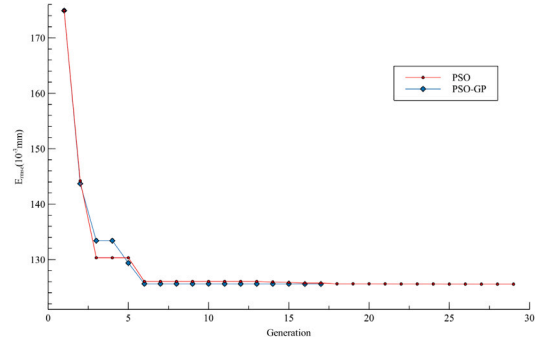


Fig. 12. Optimization results.

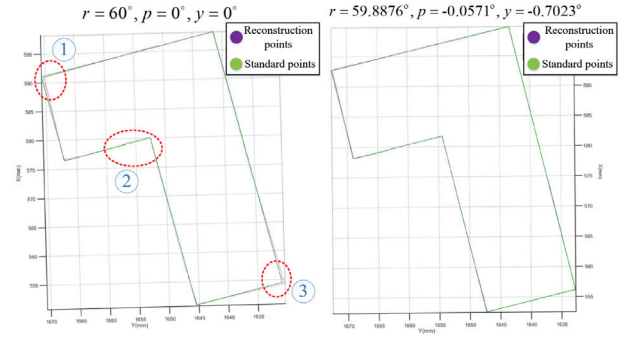


Fig. 13. Result comparison of point clouds.

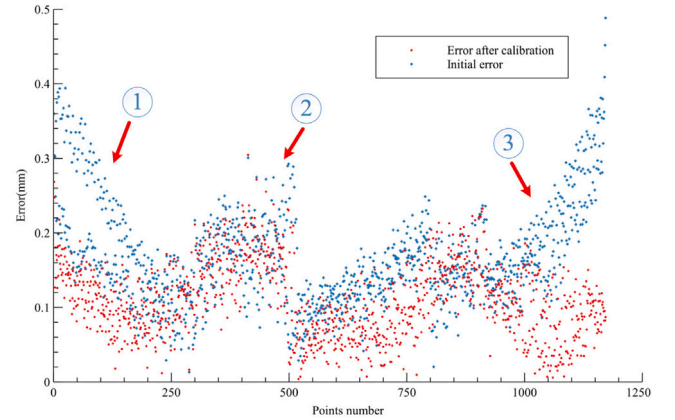


Fig. 14. Result comparison of points.

seen that after calibration, the error does drop a lot, especially in the initial situation, there are some error points greater than 0.2 mm (corresponding to large distortion part in Fig. 13). After calibration, they all reduce a vast error.

4.2.4. Calibration of translation component and comparison with traditional hand-eye calibration methods

This part mainly includes two contents. (1) The translation component is calculated based on the accurately calibrated rotation component values of hand-eye transformation. (2) Compare and analyze the method in this paper with the traditional hand-eye calibration methods.

In the experiment, the robot scans the calibration object in 6 different poses **A**, and the reconstructed point cloud is registered with the standard point cloud to obtain **B**. The results are shown in Table 2. In this paper, t_x, t_y, t_z are represented the three components of the translation part of the calibration results. After the rotation component

Table 2
Data of the translation component calibration.

Item	tx, ty, tz (mm)	r, p, y (°)
A_1	-216.56, 1604.47, 1120.96	137.75, 19.91, 176.45
A_2	-7.47, 1436.89, 1182.8	142.75, -12.18, 179.37
A_3	219.15, 1553.4, 1132.07	-143.53, -2.84, 168.15
A_4	285.08, 1573.68, 1066.17	-120.22, 8.28, 161.99
A_5	-91.67, 1601.53, 1103.15	129.01, 4.52, -167.31
A_6	-133.55, 1770.43, 1074.75	112.35, 24.36, -164.88
B_1	-5.64, 61.48, 333.31	-68.99, 22.48, -88.87
B_2	9.38, 68.90, 395.79	-62.73, -9.78, -86.77
B_3	-48.90, 30.28, 365.72	9.53, -9.98, -87.08
B_4	-17.55, 26.59, 325.14	33.02, -16.17, -81.24
B_5	16.88, 71.78, 246.98	-76.15, 3.11, -102.19
B_6	-4.03, 61.22, 234.66	-99.09, 13.52, -115.23

Table 3
Results using traditional methods.

Method	tx, ty, tz (mm)	r, p, y (°)
Andreff [8]	99.48, 181.35, 102.61	59.22, -0.70, -0.25
Chou [9]	99.09, 179.33, 95.34	58.67, -0.20, -0.28
Horaud [10]	98.83, 179.93, 94.91	58.68, -0.66, 0.22
Park [11]	99.16, 179.33, 95.44	58.70, -0.18, -0.34
Shiu [12]	98.36, 180.25, 93.51	59.33, 0.01, 0.48
Tsai [13]	98.73, 179.69, 94.78	58.60, -0.51, 0.17

Table 4
Results of the comparison.

Method	Traditional error	Recon error (mm)
Andreff	0.0275	0.2348
Chou	0.0267	0.2112
Horaud	0.028	0.2982
Park	0.0268	0.2056
Shiu	0.0281	0.2266
Tsai	0.0275	0.2781
3D Recon	0.029	0.1256

is known as $r = 59.8876^\circ, p = -0.0571^\circ, y = -0.7023^\circ$, the translation component can be obtained as $tx = 100.28$ mm, $ty = 180.51$ mm, $tz = 96.55$ mm by the proposed method in this paper.

As a comparison, six traditional hand-eye calibration methods were used, and the results are shown in Table 3.

The rotation error-index of the traditional hand-eye calibration method in Eq. (32) and the error-index of 3D reconstruction are used to compare the results of different methods, as shown in Table 4. In the table, 3D Recon means the method in this paper, Recon error means the E_{rmse} in this paper.

$$E_R = \frac{1}{n} \sum_{i=1}^n \left\| \log \left(\left(R_{X_{solve}} R_{Bi} \right) \left(R_{Ai} R_{X_{solve}} \right)^{-1} \right) \right\|. \quad (32)$$

It can be seen from the table that the traditional error-index is not positively related to the 3D reconstruction error-index, and this is because that the traditional error-index measures the ability of the solution X to meet each data A and B . It essentially measures the accuracy of the solution X rather than the reconstruction accuracy. Therefore, it can be concluded that, from the perspective of 3D reconstruction, the traditional methods are not as accurate and intuitive as the method in this paper in terms of solution method and evaluation index when solving the rotation component.

Besides, the reprojection error is also used for evaluation. Reprojection poses can be obtained from $Y_i = A_i X_{solve} B_i$ and the standard deviation of 6 reprojection positions is used as the evaluation index. The result is shown in Table 5. It can be seen from the results that the method in this paper improved some accuracy compared to other methods, and this is because a relatively accurate value of the rotation component was obtained.

Table 5
Results of the comparison using reprojection error.

Method	X (mm)	Y (mm)	Z (mm)	Distance (mm)
Andreff	2.47	3.95	1.61	4.93
Chou	2.20	1.92	1.10	3.12
Horaud	4.43	1.88	1.22	4.97
Park	2.03	2.02	1.11	3.07
Shiu	2.65	1.39	0.57	3.05
Tsai	3.96	1.68	1.13	4.45
3D Recon	0.82	2.63	1.22	3.01

5. Discussion

Comparing the experiment results, from the perspective of 3D reconstruction, the advantage of the proposed method is obvious. In the solution of the translation component, the proposed method has less error than the traditional methods, which proves that the value of the rotation component is more correct than the solutions of the traditional methods.

In addition to the above comparisons, the method in this paper is also compared with two state-of-art methods. In fact, because the traditional methods are mainly for solving the hand eye equation, the experimental data obtained in the translation component experiment of this paper can be used for comparison. For the novel methods for the hand-eye calibration of line laser sensors in recent years, the platforms and indicators are usually not the same, so they are discussed in this section.

Paper [29] proposed a similar idea to this paper. They used the method of reconstructing point cloud matching to calibrate the hand-eye relationship. In the specific method, they moved the robot to two different random poses, and rotated the end axis 180 degrees in each pose, so that the lidar sensor can obtain the point cloud of the surrounding environment, and the point cloud obtained in the two poses are used to iteratively calculate the hand-eye relationship by ICP method. But their method is aimed at two-dimensional lidar sensor, and its measurement range and distance are much larger than the line laser sensor in this paper, and the accuracy is also much lower than that of the line laser sensor. Therefore, if their method is directly used for the line laser sensor, due to the limited measurement range, it will not be able to obtain two point clouds that can be used for registration. However, it can be found that the idea of using three-dimensional reconstruction has indeed achieved successful applications in robot hand-eye calibration.

Paper [28] proposed a method of using a plane artifact to calibrate the line laser hand-eye relationship. They conducted a lot of simulation research to analyze the relationship between calibration error and various factors, and finally used the error index of the fitting plane as the evaluation standard of hand-eye calibration in the experiment. However, the method in our paper limits the robot pose in the measurement, which is aiming at reducing the absolute error of the laser sensor, so the same measurement data cannot be obtained to directly compare the two methods. However, in comparison, the new index proposed in this paper more intuitively reflects the accuracy of reconstruction. At the same time, the method in this paper directly reflects the accuracy of hand-eye calibration to the measurement task.

In the analysis of the error of the results of this paper, it can be found that there is still an error of about 0.1 mm. Meanwhile, this error slows the convergence of the optimization algorithm. There are the following kinds of errors, including: robot positioning accuracy, sensor variation error at different measuring distances, the error caused by the robot movement interval and the sensor measuring point interval, and the size error of the calibration plate. But in general, this measurement error can meet most of the robotic industrial applications.

6. Conclusion

In this paper, the author first analyzed the influence of the line laser sensor hand-eye relationship error on the absolute and relative errors of the 3D reconstruction of the point cloud. It is concluded that when the robot performs a translation scanning, the rotation component of the hand-eye relationship can be calibrated by optimizing the errors of the 3D reconstruction. Therefore, the paper proposed a hand-eye calibration method and evaluation index based on 3D reconstruction. PSO-GP method was used as the optimization method to improve the calibration performance. Meanwhile, the paper presented a calibration method of the translation component of the hand-eye relationship, which was achieved by constructing the traditional hand-eye calibration matrix $AX = XB$. In the end, simulation experiments were conducted to verify the effectiveness of the calibration method and optimization method for calibrating the rotation component. At the same time, real experiments were performed to verify the effectiveness of the proposed calibration method, and the final reconstruction accuracy was 0.1256 mm. The method in this paper is compared with the traditional methods, and it is concluded that the error-index of the traditional method is not reasonable when analyzed from the perspective of 3D reconstruction. Besides, when the rotation component value obtained by optimizing the 3D reconstruction error is used to calculate the overall hand-eye matrix, its reprojection error is better than those of traditional methods. In the future, the research includes selecting better experimental configurations to improve calibration accuracy, and the translation component of the hand eye calibration will be tried to achieve by the 3D reconstruction method. In summary, the calibration results can meet most of the measurement needs. Compared with traditional methods, the method proposed in this paper is more convenient and accurate, and the evaluation indicators are more intuitive.

CRedit authorship contribution statement

Mingyang Li: Conceptualization, Methodology, Writing - original draft, Writing - review & editing. **Zhijiang Du:** Supervision, Validation. **Xiaoxing Ma:** Writing - review & editing. **Wei Dong:** Resources, Writing - review & editing. **Yongzhuo Gao:** Software, Investigation, Formal analysis, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work is supported by the National Science and Technology Major Project of China under Grant No. 2017ZX04005001-005, the Joint Funds of the National Natural Science Foundation of China under Grant U1813213, and the National Natural Science Foundation of China under Grants 61773141.

References

- [1] Chengyi Yu, Juntong Xi, Simultaneous and on-line calibration of a robot-based inspecting system, *Robot. Comput.-Integr. Manuf.* 49 (2017) (2018) 349–360.
- [2] Dahu Zhu, Xiaozhi Feng, Xiaohu Xu, Zeyuan Yang, Wenlong Li, Sijie Yan, Han Ding, Robotic grinding of complex components: A step towards efficient and intelligent machining – challenges, solutions, and applications, *Robot. Comput.-Integr. Manuf.* 65 (2019) (2020) 101908.
- [3] B.W. He, Zhou Xiao Long, Y.F. Li, The research of an automatic object reconstruction method based on limit visible region of the laser-scanning vision system, *Robot. Comput.-Integr. Manuf.* 26 (6) (2010) 711–719, 19th International Conference on Flexible Automation and Intelligent Manufacturing.
- [4] Wen Long Li, He Xie, Gang Zhang, Si Jie Yan, Zhou Ping Yin, 3-D shape matching of a blade surface in robotic grinding applications, *IEEE/ASME Trans. Mechatronics* 21 (5) (2016) 2294–2306.
- [5] Vincenzo Niola, Cesare Rossi, Sergio Savino, Salvatore Strano, A method for the calibration of a 3-D laser scanner, *Robot. Comput.-Integr. Manuf.* 27 (2) (2011) 479–484.
- [6] Shibin Yin, Yongjie Ren, Yin Guo, Jigui Zhu, Shourui Yang, Shenghua Ye, Development and calibration of an integrated 3D scanning system for high-accuracy large-scale metrology, *Meas.: J. Int. Meas. Confed.* 54 (2014) 65–76.
- [7] Wen Long Li, He Xie, Gang Zhang, Si Jie Yan, Zhou Ping Yin, Hand-eye calibration in visually-guided robot grinding, *IEEE Trans. Cybern.* 46 (11) (2016) 2634–2642.
- [8] N. Andreff, R. Horaud, B. Espiau, On-line hand-eye calibration, in: *Second International Conference on 3-D Digital Imaging and Modeling (Cat. No.PR00062)*, 1999, pp. 430–436.
- [9] Jack C.K. Chou, M. Kamel, Finding the position and orientation of a sensor on a robot manipulator using quaternions, *Int. J. Robot. Res.* 10 (3) (1991) 240–254.
- [10] Radu Horaud, Fadi Dornaika, Hand-eye calibration, *Int. J. Robot. Res.* 14 (3) (1995) 195–210.
- [11] F.C. Park, B.J. Martin, Robot sensor calibration: solving $AX=XB$ on the Euclidean group, *IEEE Trans. Robot. Autom.* 10 (5) (1994) 717–721.
- [12] Y.C. Shiu, S. Ahmad, Calibration of wrist-mounted robotic sensors by solving homogeneous transform equations of the form $AX=XB$, *IEEE Trans. Robot. Autom.* 5 (1) (1989) 16–29.
- [13] Roger Y. Tsai, Reimar K. Lenz, et al., A new technique for fully autonomous and efficient 3 D robotics hand/eye calibration, *IEEE Trans. Robot. Autom.* 5 (3) (1989) 345–358.
- [14] Jan Heller, Michal Havlena, Tomas Pajdla, Globally optimal hand-eye calibration using branch-and-bound, *IEEE Trans. Pattern Anal. Mach. Intell.* 38 (5) (2016) 1027–1033.
- [15] Thomas Ruland, Tomas Pajdla, Lars Kruger, Globally optimal hand-eye calibration, in: *Proceedings of the IEEE Computer Society Conference on Computer Vision and Pattern Recognition*, (2) (2012), pp. 1035–1042.
- [16] Z. Zhao, Hand-eye calibration using convex optimization, in: *2011 IEEE International Conference on Robotics and Automation*, 2011, pp. 2947–2952.
- [17] Ihtisham Ali, Olli Suominen, Atanas Gotchev, Emilio Ruiz Morales, Methods for simultaneous robot-world-hand-eye calibration: A comparative study, *Sensors (Switzerland)* 19 (12) (2019).
- [18] M.K. Ackerman, A. Cheng, G. Chirikjian, An information-theoretic approach to the correspondence-free $AX=XB$ sensor calibration problem, in: *2014 IEEE International Conference on Robotics and Automation (ICRA)*, 2014, pp. 4893–4899.
- [19] Qianli Ma, Haiyuan Li, G.S. Chirikjian, New probabilistic approaches to the $AX = XB$ hand-eye calibration without correspondence, in: *2016 IEEE International Conference on Robotics and Automation (ICRA)*, 2016, pp. 4365–4371.
- [20] Huajian Song, Zhijiang Du, Weidong Wang, Lining Sun, Singularity analysis for the existing closed-form solutions of the hand-eye calibration, *IEEE Access* 6 (2018) 75407–75421.
- [21] Huy Nguyen, Quang Cuong Pham, On the covariance of X in $AX = XB$, *IEEE Trans. Robot.* 34 (6) (2017) 1651–1658.
- [22] Xiaohu Xu, Dahu Zhu, Haiyang Zhang, Sijie Yan, Han Ding, TCP-based calibration in robot-assisted belt grinding of aero-engine blades using scanner measurements, *Int. J. Adv. Manuf. Technol.* 90 (1) (2017) 635–647.
- [23] Defeng Wu, Tianfei Chen, Aiguo Li, A high precision approach to calibrate a structured light vision sensor in a robot-based three-dimensional measurement system, *Sensors (Switzerland)* 16 (9) (2016).
- [24] Wenyu Chen, Jia Du, Wei Xiong, Yue Wang, Shueching Chia, Bingbing Liu, Jierong Cheng, Ying Gu, A noise-tolerant algorithm for robot-sensor calibration using a planar disk of arbitrary 3-D orientation, *IEEE Trans. Autom. Sci. Eng.* 15 (1) (2018) 251–263.
- [25] Shibin Yin, Yongjie Ren, Jigui Zhu, Shourui Yang, Shenghua Ye, A vision-based self-calibration method for robotic visual inspection systems, *Sensors (Switzerland)* 13 (12) (2013) 16565–16582.
- [26] Jorge Santolaria, Juan José Aguilar, David Guillomía, Carlos Cajal, A crenellated-target-based calibration method for laser triangulation sensors integration in articulated measurement arms, *Robot. Comput.-Integr. Manuf.* 27 (2) (2011) 282–291.
- [27] Fredrik Bagge Carlson, Rolf Johansson, Anders Robertsson, Six DOF eye-to-hand calibration from 2D measurements using planar constraints, in: *2015 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, IEEE, 2015, pp. 3628–3632.
- [28] Sara Sharifzadeh, Istvan Biro, Peter Kinnell, Robust hand-eye calibration of 2D laser sensors using a single-plane calibration artefact, *Robot. Comput.-Integr. Manuf.* 61 (2019) (2020) 101823.
- [29] Arne Peters, Adam Schmidt, Alois C. Knoll, Extrinsic calibration of an eye-in-hand 2D LiDAR sensor in unstructured environments using ICP, *IEEE Robot. Autom. Lett.* 5 (2) (2020) 929–936.
- [30] Micro epsilon Company, Scancontrol 29x0 product index, 2020, http://www.micro-epsilon.com.cn/a/chanpinzhongxin/2D_3Djiguangsaomiaoyi/561.html/, Accessed March 18.

- [31] Paul J. Besl, Neil D. McKay, A method for registration of 3-D shapes, *IEEE Trans. Pattern Anal. Mach. Intell.* 14 (2) (1992) 239–256.
- [32] Jiaolong Yang, Hongdong Li, Yunde Jia, Go-ICP: Solving 3D registration efficiently and globally optimally, in: *Proceedings of the IEEE International Conference on Computer Vision*, (2013), pp. 1457–1464.
- [33] James Kennedy, Russell Eberhart, 47-particle swarm optimization proceedings, in: *IEEE International Conference*, in: *Proceedings of ICNN'95 - International Conference on Neural Networks*, vol. 11, 1995, pp. 111–117, (1).
- [34] Matthias Seeger, Gaussian processes for machine learning University of California at Berkeley, *Int. J. Neural Syst.* 14 (2004) 69–109.